

## Chapter 16: Abstract Vector Spaces

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In Chapters 12 through 14, we worked algebraically with vectors in  $\mathbb{R}^n$ . Many of our results relied on the fact that vectors in  $\mathbb{R}^n$  can be added and multiplied by scalars in a familiar way.

However, vectors in  $\mathbb{R}^n$  is not the only set that has these properties. Other examples of sets that satisfy the same properties include: polynomials, differentiable functions, matrices, and others! Since so many useful sets exhibit the same properties, we will develop a name and theory surrounding these sets with addition and scalar multiplication that “play nice” with each other.

*Definition:* A (real) **vector space** consists of a nonempty set  $V$  of objects (called **vectors**) that can be added, that can be multiplied by a real number (called a **scalar** in this context), and for which certain axioms hold. If  $\mathbf{v}$  and  $\mathbf{w}$  are two vectors in  $V$ , their sum is expressed as  $\mathbf{v} + \mathbf{w}$ , and the scalar product of  $\mathbf{v}$  by a real number  $a$  is denoted as  $a\mathbf{v}$ . The following axioms are assumed to hold for a vector space.

- A1. If  $\mathbf{u}$  and  $\mathbf{v}$  are any elements in  $V$ , then  $\mathbf{u} + \mathbf{v}$  is in  $V$ . In this case, we say  $V$  is **closed** under vector addition.
- A2.  $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$  for all  $\mathbf{u}, \mathbf{v}$  in  $V$ .
- A3.  $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$  for all  $\mathbf{u}, \mathbf{v}, \mathbf{w}$  in  $V$ .
- A4. There exists an element  $\mathbf{0}$  (written \_\_\_\_\_) in  $V$  such that  $\mathbf{v} + \mathbf{0} = \mathbf{0} + \mathbf{v} = \mathbf{v}$  for any  $\mathbf{v} \in V$ . The vector  $\mathbf{0}$  is called the **zero vector**.
- A5. For each  $\mathbf{v}$  in  $V$  there exists an element  $-\mathbf{v}$  in  $V$  such that  $\mathbf{v} + (-\mathbf{v}) = -\mathbf{v} + \mathbf{v} = \mathbf{0}$ . The vector  $-\mathbf{v}$  is called the **negative** of  $\mathbf{v}$  or **additive inverse** of  $\mathbf{v}$ .
- S1. If  $\mathbf{v}$  is any element in  $V$  and  $a$  is any real number, then  $a\mathbf{v}$  is in  $V$ . In this case, we say  $V$  is **closed** under scalar multiplication.
- S2.  $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$  for any  $\mathbf{u}, \mathbf{v}$  in  $V$  and any  $a$  in  $\mathbb{R}$ .
- S3.  $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$  for any  $\mathbf{v}$  in  $V$  and any  $a, b$  in  $\mathbb{R}$ .
- S4.  $a(b\mathbf{v}) = (ab)\mathbf{v}$  for any  $\mathbf{v}$  in  $V$  and any  $a, b$  in  $\mathbb{R}$ .
- S5.  $1\mathbf{v} = \mathbf{v}$  for any  $\mathbf{v}$  in  $V$ .

The operation  $\mathbf{v} + \mathbf{w}$  is called **vector addition** and, and the operation  $av$  is called **scalar multiplication**.

*Note:* When using a “non-standard” vector addition and/or scalar multiplication, we will denote vector addition as  $\oplus$  and scalar multiplication as  $\odot$ .

**Example:** Let  $V$  be the set of all positive real numbers, and define vector addition  $\oplus$  by  $u \oplus v = uv$  ( $\oplus$  is ordinary multiplication) and define scalar multiplication  $\odot$  by  $c \odot v = v^c$ . What is the zero vector  $\mathbf{0}$  for  $V$ ?

$$\vec{0} = 1.$$

(Note: you will show the rest of the vector space axioms are true for this  $(V, \oplus, \odot)$  in discussion, so this is indeed a vector space!)

*Note:* We will often just use the phrase “vector space,” dropping the “real” and assuming our scalars are from the real numbers in this course. However, it is possible to have complex vector spaces, rational vector spaces, etc.!

*Warning!* From now on, “vector” will just mean an element of a vector space, not necessarily a directed line segment.

**Example:** The set  $\mathbb{R}^n$  is a real vector space for any natural number  $n$ . This is the most common “day-to-day” vector space.

**Example:** The set of all  $m \times n$  matrices with real entries with matrix addition as addition and scalar multiplication of a matrix by a real number as scalar multiplication is a vector space. We denote this vector space by  $M_{mn}$ ,  $M_{m \times n}$ , or  $\mathbb{R}^{m \times n}$ . Prove that  $M_{2 \times 2}$  is closed under addition and scalar multiplication. What is the zero vector in this vector space?

$$\forall A, B \in M_{m \times n}, A + B \in M_{m \times n}$$

$$\forall c \in \mathbb{R}, c \cdot A \in M_{m \times n}.$$

so addition and scalar mult. are closed.

zero vector is  $0_{m \times n}$  matrix.

**Example:** A **polynomial** (in  $x$ ) is a function expressible as

$$p(x) = a_0 + a_1x + \cdots + a_dx^d.$$

where  $a_0, a_1, \dots, a_d$  are real numbers and  $d$  is a nonnegative integer. If  $a_d \neq 0$ , we say  $p(x)$  has **degree**  $d$ .

Let  $P(\mathbb{R})$  or  $P$  be the set of all polynomials (with real coefficients), and let  $P_d(\mathbb{R})$  or  $P_d$  be the set of all polynomials (with real coefficients) of degree  $\leq d$  together with the zero polynomial  $p(x) = 0$ , denoted  $\mathbf{0}$ .

Using the standard polynomial addition and scalar multiplication for polynomials,  $P$  is a vector space and each  $P_d$  is a vector space.

Let's walk through the proof of some of the axioms that  $P_2$  (the set of polynomials of degree at most 2) is a vector space. We list them all here for completeness, but we'll choose a few to do in class.

(A1) If  $\mathbf{u}$  and  $\mathbf{v}$  are in  $V$ , then  $\mathbf{u} + \mathbf{v}$  is in  $V$ .

$$\begin{aligned} \mathbf{u}(x) &= a_0 + a_1x + \cdots + a_dx^d \\ \mathbf{v}(x) &= b_0 + b_1x + \cdots + b_dx^d \\ \Rightarrow \mathbf{u} + \mathbf{v} &= (a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_d + b_d)x^d \in P_d(\mathbb{R}) \end{aligned}$$

(A2)  $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$  for all  $\mathbf{u}, \mathbf{v}$  in  $V$ .

(A3)  $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$  for all  $\mathbf{u}, \mathbf{v}, \mathbf{w}$  in  $V$ .

(A4) There exists an element  $\mathbf{0}$  in  $V$  such that  $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$  for all  $\mathbf{u}$  in  $V$ .

$\vec{0}$  is the  $0$  polynomial.

(A5) For each  $\mathbf{u}$  in  $V$  there exists an element  $-\mathbf{u}$  in  $V$  such that  $\mathbf{u} + -\mathbf{u} = -\mathbf{u} + \mathbf{u} = \mathbf{0}$ .

$$-\mathbf{u} = -a_0 - a_1x - \cdots - a_dx^d.$$

(S1) If  $\mathbf{v}$  is in  $V$  and  $r$  is in  $\mathbb{R}$ , then  $r\mathbf{v}$  is in  $V$ .

$$r\mathbf{v} = rb_0 + rb_1x + \cdots + rb_dx^d.$$

(S2)  $r(\mathbf{u} + \mathbf{v}) = r\mathbf{u} + r\mathbf{v}$  for all  $\mathbf{u}, \mathbf{v}$  in  $V$  and  $r$  in  $\mathbb{R}$ .

(S3)  $(r + s)\mathbf{v} = r\mathbf{v} + s\mathbf{v}$  for all  $\mathbf{v}$  in  $V$  and  $r, s$  in  $\mathbb{R}$ .

(S4)  $r(s\mathbf{v}) = (rs)\mathbf{v}$  for all  $\mathbf{v}$  in  $V$  and  $r, s$  in  $\mathbb{R}$ .

(S5)  $1\mathbf{v} = \mathbf{v}$  for all  $\mathbf{v}$  in  $V$ .

$$1 \cdot (b_0 + b_1x + \cdots + b_dx^d) = b_0 + b_1x + \cdots + b_dx^d.$$

**Example:** Let  $V$  be the set of all ordered pairs of real numbers  $(x, y)$  with the operations

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1, 0) \text{ and } c \odot (x, y) = (cx, cy).$$

Show that  $V$  is not a vector space.

NOT exist zero vector!

$$\text{Let } \vec{0} = (x_0, y_0), \text{ then } (x_0, y_0) \oplus (x_2, y_2) = (x_2, y_2) \neq (x_2, \vec{0}).$$

**Example:** Let  $V$  be the set of all ordered pairs of real numbers  $(x, y)$  with the operations

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1 + x_2, y_1 + y_2) \text{ and } c \odot (x, y) = (cx, y).$$

Show that  $V$  is not a vector space.

$$0 \odot (x, y) = (0, 0) \neq (0, y).$$

So NOT a vector space.

**Example:** Consider  $V =$  non-negative real numbers, and define vector addition as  $\mathbf{v} \oplus \mathbf{u} = |\mathbf{v} - \mathbf{u}|$ , and scalar multiplication as  $a \odot \mathbf{v} = |a|\mathbf{v}$ . Does this form a vector space?

$$\text{Let } v = 3, u = 2, w = 5$$

$$(v \oplus u) \oplus w = 1 \oplus 5 = 4$$

$$v \oplus (u \oplus w) = 3 \oplus 3 = 0$$

Does NOT satisfy associativity.

Below are some useful facts about the interaction of addition and scalar multiplication with the zero vector and the scalar zero.

**Theorem:** Let  $V$  be a vector space. For any  $\mathbf{u}$  in  $V$  and  $a$  in  $\mathbb{R}$ , then

- (a)  $0\mathbf{u} = \mathbf{0}$
- (b)  $a\mathbf{0} = \mathbf{0}$
- (c) If  $a\mathbf{u} = \mathbf{0}$ , then  $a = 0$  or  $\mathbf{u} = \mathbf{0}$ .
- (d)  $(-1)\mathbf{u} = -\mathbf{u}$
- (e)  $(-a)\mathbf{u} = -(a\mathbf{u}) = a(-\mathbf{u})$ .

Proof of \_\_\_\_\_ (Pick one):

$$(a) \quad 0 \cdot \vec{u} = (a - a) \cdot \vec{u} = a \cdot \vec{u} - a \cdot \vec{u} = \vec{0}.$$

$$(b) \quad a \cdot \vec{0} = a \cdot (\vec{u} - \vec{u}) = a \cdot \vec{u} - a \cdot \vec{u} = \vec{0}.$$

$$(c) \quad \text{If } a \neq 0 \text{ and } \vec{u} \neq \vec{0}, \text{ then } \vec{0} \neq \vec{u} = 1 \cdot \vec{u} \\ = \left(\frac{1}{a} \cdot a\right) \cdot \vec{u} \\ = \frac{1}{a} \cdot (a \cdot \vec{u}) = \frac{1}{a} \cdot \vec{0}$$